

Ella Johnson

Madison, CT • emj67@cornell.edu • (808) 688-4942
<https://www.linkedin.com/in/ellamjohnson/>
<https://cornell-mae-ug.github.io/fa25-portfolio-ellabellaboo54/>

EDUCATION

Cornell University Ithaca, NY
Bachelor of Science, Mechanical Engineering, **Minor** Theater Expected May 2027

Honors: DoW SMART Scholar, Cornell University Dean's List

Courses: Introduction to Mechanical Design, Innovative Product Design, Introductory Fluid Mechanics, Mechatronics, Mechanics of Engineering Materials, Introduction to Aeronautics

EXPERIENCE

Cornell Duffield College of Engineering Ithaca, NY
Peer Advisor August 2026 - Present

- Plan and lead weekly advising seminars covering ways to be successful at college.
- Provide guidance and assistance with the transition to college life, including study skills, class recommendations, and ways to get involved.

General Dynamics Electric Boat New London, CT
Intern - Design and Engineering Summer 2025, Winter 2026, Summer 2026

- Developed and revised technical calculation reports verifying the adequacy and compliance of engineering designs of new Virginia-class submarine hull and non-pressure hull structural designs for stakeholder approval.
- Performed finite element analysis to determine structural adequacy of design.
- Facilitated effective kickoff and follow-up meetings that established project goals and methods, tracked progress, and maintained stakeholder alignment.
- Reviewed and updated engineering drawings to incorporate changes in requirements and designs for future ships.
- Audited vendor-provided information for acceptance of manufactured parts to maintain quality standards and ensure requirements were met.

Cornell Mechanical and Aerospace Engineering Ithaca, NY
Intro to Mechanical Design TA January - May 2026

- Taught weekly lab sections and held office hours to provide assistance to students that needed clarification of class concepts.
- Assisted students with CAD fundamentals to ensure they were able to complete class assignments.

Soundview Family YMCA Branford, CT
Swim Instructor, Lifeguard, Membership Services Representative January - August 2024

- Instructed group and private lessons to swimmers ages 3 to 12 to develop skills and confidence in water. Adapted lesson plans to the needs and temperaments of the students while maintaining curriculum requirements.
- Monitored and ensured safety of patrons in water and on pool deck by administering first aid and responding promptly to emergencies. Communicated clearly and effectively to patrons pool rules, safety protocols, and emergency procedures to maintain order and safety.
- Acted as an ambassador of all YMCA programs with a focus on membership and member engagement.

ACTIVITIES

Cornell Big Red Pep and Marching Bands Ithaca, NY
Trombone Section Leader (2025), Pep Social Chair, (202X) Pep Conductor (2026) September 2022 - Present

- Perform at weekly rehearsals and university athletic events, contributing 10–20 hours per week to support team spirit and band performance
- Lead section activities, assist with new member recruitment, onboard and train on music, drills, and band procedures.
- Plan and coordinate social events by securing venues, arranging food, managing required documentation, and communicating logistics with band members.

SKILLS

Adaptability • Technical Writing • Leadership • Collaboration • Autodesk Fusion 360